

STACK QNO:1

QUESTION:

Problem Description

A and B are playing a game. In this game, both of them are initially provided with a **list of n numbers**. (Both have the same list but their own copy).

Now, they both have a different strategy to play the game. A picks the element from **start of his list**. B picks from the **end of his list**.

You need to generate the result in form of an output list.

Method to be followed at each step to build the output list is:

1. If the number picked by A is **bigger than** B then this step's **output is 1** . B **removes** the number that was picked from their list.
2. If the number picked by A is **smaller than** B then this step's **output is 2** . A **removes** the number that was picked from their list.
3. If both have the **same number** then this step's **output is 0** . Both A and B **remove** the number that was picked from their list.

This game **ends** when at least one of them has no more elements to be picked i.e. when the **list gets empty**.

Output the built output list.

Constraints

$$1 \leq N \leq 10^6$$

$$1 \leq \text{num} \leq 10^9$$

Input format:

First line consists of a number n , size of the list provided.

Next line consists of n numbers separated by space.

Output format:

Output the required output list.

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 3 1 2 3 EXPECTED OUTPUT 2 2 0	INPUT (STDIN) 5 5 7 1 2 3 EXPECTED OUTPUT 1 1 1 2 0 2 2 2

▼ Mandatory Test Cases

Test Case 1
KEYWORD if(a[i]>a[j])

▼ Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY 6	TOKEN COUNT 160	NLOC 36

Source Code:

```
#include <stdio.h>
int a[1000005];
int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    for (int k = 0; k < n; k++) {
        scanf("%d", &a[k]);
    }

    int i = 0;
    int j = n - 1;

    while (i < n && j >= 0) {
        if(a[i]>a[j]) {
            printf("1 ");
            j--;
        }
        else if (a[i] < a[j]) {
            printf("2 ");
            i++;
        }
        else {
            printf("0 ");
            i++;
            j--;
        }
    }

    printf("\n");
    return 0;
}
```

Output:

```
3
1 2 3
2 2 0
```

```
5
5 7 1 2 3
1 1 1 2 0 2 2 2
```